

Inner Product Space

Consider a vector space $\mathbb{R}^n$.
Prove that $\mathbb{R}^n$ is an
Inner Product Space, which

means $\rightarrow$

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i \rightarrow \text{Standard Inner Product}$$

★ Is it a Binary Operator :- $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ Therefore, it is not Binary.

Condition for Inner Product Space :- ① Linearity : $\langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
and $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$

This is equivalent to :- $\langle \alpha u + \beta v, w \rangle = \alpha \langle u, w \rangle + \beta \langle v, w \rangle$

Let u, v, w be vectors in vector space $\mathbb{R}^n$.
 $u = [u_1, u_2, \dots, u_n]^T$, $v = [v_1, v_2, \dots, v_n]^T$ & $w = [w_1, w_2, \dots, w_n]^T$
Here u_i, v_i & $w_i \in \mathbb{R}$

$$\langle u, w \rangle = \sum_{i=1}^n u_i w_i \quad \& \quad \langle v, w \rangle = \sum_{i=1}^n v_i w_i ; \quad \langle u+v, w \rangle = \sum_{i=1}^n (u_i + v_i) w_i$$

$$= \sum_{i=1}^n u_i w_i + \sum_{i=1}^n v_i w_i$$

$$\therefore \langle u+v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$$

$$\text{Now, } \alpha u = [\alpha u_1, \alpha u_2, \dots, \alpha u_n] ; \quad \langle \alpha u, v \rangle = \sum_{i=1}^n \alpha u_i v_i = \alpha \sum_{i=1}^n u_i v_i$$

$$\text{Thus, } \langle \alpha u, v \rangle = \alpha \langle u, v \rangle$$

So, $\langle u, v \rangle$ is Linear.

② Symmetry $\rightarrow \langle u, v \rangle = \langle v, u \rangle$ For each question $\rightarrow$

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i = \sum_{i=1}^n v_i u_i = \langle v, u \rangle$$

Since real no.s are Commutative

③ Positive Definiteness - $\langle \underline{u}, \underline{u} \rangle \geq 0$; $\langle \underline{u}, \underline{u} \rangle = \sum_{i=1}^n u_i^2 \geq 0$
It is 0 only if $\underline{u} = 0$

Example 1- $\underline{u} = \begin{bmatrix} 2+3i \\ 5 \end{bmatrix}$ & $\underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}$; $\langle \underline{u}, \underline{v} \rangle = \sum_{i=1}^2 u_i v_i = 6i - 9 + 25 = 19 + 6i$
 $\langle \underline{v}, \underline{u} \rangle = \langle \underline{u}, \underline{v} \rangle = 19 + 6i$

Example -2 $\rightarrow \underline{u} = \begin{bmatrix} 2+2i \\ 3 \end{bmatrix}$; $\underline{v} = \begin{bmatrix} 3i \\ 5 \end{bmatrix}$; $\underline{w} = \begin{bmatrix} 3 \\ 5i \end{bmatrix}$
 $\langle \underline{u}, \underline{u} \rangle = 4 - 4 + 8i + 9$; $\langle \underline{v}, \underline{v} \rangle = -9 + 25$; $\langle \underline{w}, \underline{w} \rangle = 9 - 25$

Positive Definiteness
not satisfied $\leftarrow$

Standard Inner Product

$$- \langle u, v \rangle = \sum_{i=1}^n u_i \overline{v_i}$$

$$\langle \underline{u}, \underline{u} \rangle = (2+2i)(2-2i) + 9 = 17 ; \langle \underline{v}, \underline{v} \rangle = 9+25 = 34$$

$$\langle \underline{w}, \underline{w} \rangle = 9+25 = 34 ; \text{ In this case, the third condition is satisfied.}$$

General Condition :-

- ① Linearity - $\langle \alpha \underline{u} + \beta \underline{v}, \underline{w} \rangle = \alpha \langle \underline{u}, \underline{w} \rangle + \beta \langle \underline{v}, \underline{w} \rangle$
- ② Symmetry - $\langle \underline{u}, \underline{v} \rangle = \overline{\langle \underline{v}, \underline{u} \rangle}$
- ③ Positive Definiteness - $\langle \underline{u}, \underline{u} \rangle \geq 0$ Note :- $\langle \underline{u}, \alpha \underline{v} \rangle = \overline{\langle \alpha \underline{v}, \underline{u} \rangle} = \overline{\alpha \langle \underline{v}, \underline{u} \rangle} = \overline{\alpha} \overline{\langle \underline{v}, \underline{u} \rangle} = \overline{\alpha} \langle \underline{u}, \underline{v} \rangle$

Qp.1) Determine the value of λ which makes the following operation an inner product $\rightarrow \underline{x}, \underline{y} \in \mathbb{R}^2 ; \langle \underline{x}, \underline{y} \rangle = x_1 y_1 - 3x_1 y_2 - 3x_2 y_1 + \lambda x_2 y_2$

For the above to be an I.P. $\rightarrow \langle \underline{x}, \underline{x} \rangle \geq 0 \Rightarrow x_1^2 - 3x_1 x_2 - 3x_2 x_1 + \lambda x_2^2 \geq 0$

$$\therefore x_1^2 - 6x_1 x_2 + \lambda x_2^2 \geq 0 \quad \equiv y^2 - 6y + \lambda \geq 0 \quad [y = \frac{x_1}{x_2}]$$

This will be true if $b^2 - 4ac \leq 0 \Rightarrow 36 - 4\lambda \leq 0 \Rightarrow \lambda \geq 9$

Qp.2) Determine a non-zero vector in $\mathbb{R}^3$ which is **Orthogonal** to vector $\underline{v}_1, \underline{v}_2$ & $\underline{v}_3$

$$\underline{v}_1 = [1 \ 1 \ 2]^T, \underline{v}_2 = [2 \ 1 \ 3]^T \text{ \& \> } \underline{v}_3 = [1 \ 2 \ 3]^T$$

Let $\underline{x} = [a \ b \ c]^T \in \mathbb{R} ;$ Orthogonal means $\rightarrow \langle \underline{x}, \underline{v}_1 \rangle = 0$

$$\Rightarrow a + b + 2c = 0 ; \text{ Similarly } \rightarrow 2a + b + 3c = 0 \text{ \& \> } a + 2b + 3c = 0$$

$$\therefore \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & 1 & 3 & 0 \\ 1 & 2 & 3 & 0 \end{array} \right] \leftrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \leftrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Degree of Freedom = 1 $\Rightarrow c = \alpha ; b = -\alpha \Rightarrow a = -\alpha$

$$\therefore \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = \alpha^* \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

Qp. Verify whether the following are Orthogonal $\rightarrow$

- a) $\sin(x), \cos(x)$ in $[-\pi, \pi]$
- b) $\sin(n\pi x), \sin(m\pi x)$ in $[-1, 1]$
- c) $\sin(n\pi x), \cos(n\pi x)$ in $[-1, 1]$
- d) $\cos(n\pi x), \cos(m\pi x)$ in $[-1, 1]$

$$\langle f(x), g(x) \rangle = \int_a^b f(x) \overline{g(x)} dx$$

a) $f(x) = \sin(x) \text{ \& \> } g(x) = \cos(x) \Rightarrow \langle f(x), g(x) \rangle = \frac{1}{2} \int_{-\pi}^{\pi} 2 \sin(x) \cos(x) dx$

$$= \frac{1}{2} \int_{-\pi}^{\pi} \sin(2x) dx = 0$$

$\downarrow$ odd Function

$$\begin{aligned}
 b.) \quad f(x) &= \sin(n\pi x) \quad \& \quad g(x) = \sin(m\pi x) \\
 \therefore \langle f(x), g(x) \rangle &= \int_{-1}^1 \sin(m\pi x) \sin(n\pi x) dx = \frac{1}{2} \left[\int_{-1}^1 \cos((m-n)\pi x) - \cos((m+n)\pi x) \right] \\
 &= \frac{1}{2} \left[\frac{1}{(m-n)\pi} \sin((m-n)\pi x) \Big|_{-1}^1 - \frac{1}{(m+n)\pi} \sin((m+n)\pi x) \Big|_{-1}^1 \right] \\
 &= 0 \quad (\text{For } m \neq n)
 \end{aligned}$$

$$\begin{aligned}
 \text{For } m = n \quad \rightarrow \quad \int_{-1}^1 \sin^2(n\pi x) dx &= \frac{1}{2} \int_{-1}^1 (1 - \cos(2n\pi x)) dx \\
 &= \frac{1}{2} \left[2 - \frac{\sin(2n\pi x)}{2n\pi} \Big|_{-1}^1 \right] = 1 \quad \rightarrow \quad \text{Not Orthogonal}
 \end{aligned}$$

Note :- $f(x) = \sum_{n=1}^{\infty} a_n \sin(n\pi x) \rightarrow$ Fourier Series Expansion
 Taking inner product with $\sin(m\pi x)$ from -1 to 1 :-

$$\langle f(x), \sin(m\pi x) \rangle = \int_{-1}^1 \sum_{n=1}^{\infty} a_n \sin(n\pi x) \sin(m\pi x) dx$$

Qp. Let P be the vector space of all polynomial over $\mathbb{R}$, degree less than or equal to 2. Determine a polynomial which is orthogonal to $f(t)$, if :-
 $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$ where $f(t) = 2t + 1$

$$\text{Let } g(t) = a_0 + a_1 t + a_2 t^2$$

$$\text{Now, } \int_0^1 (a_0 + a_1 t + a_2 t^2) (1 + 2t) dt = \int_0^1 (a_0 + t(a_1 + 2a_0) + t^2(a_2 + 2a_1) + 2a_2 t^3) dt$$

$$\Rightarrow a_0 + \frac{a_1 + 2a_0}{2} + \frac{2a_1 + a_2}{3} + \frac{a_2}{4} = 0$$

$$\therefore 12a_0 + 7a_1 + 5a_2 = 0 \quad \text{Degree of Freedom} = 2$$

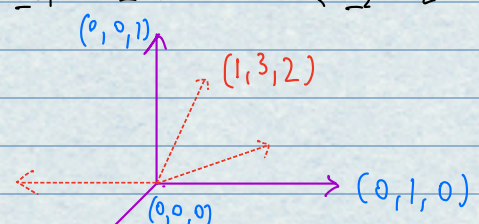
$$a_0 = \alpha \quad \& \quad a_1 = \beta \quad \Rightarrow \quad a_2 = -\frac{5}{12}\beta - \frac{7}{12}\alpha$$

$$\text{Thus, } \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ -\frac{7}{12} \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ -\frac{5}{12} \end{bmatrix}$$

Norm & Metric Spaces

Qp. 2 vectors u_1 & u_2 in a 2D vector space $\rightarrow u_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ & $u_2 = \begin{bmatrix} i & 1 \end{bmatrix}^T$

- Are u_i 's Linear Independent?
- Do they form an Orthogonal basis?
- If not, obtain an orthogonal basis from them.



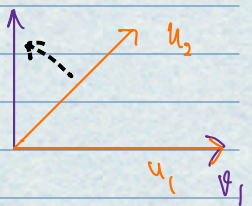
$$a) x_1 \underline{u}_1 + x_2 \underline{u}_2 = 0 \Rightarrow c_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (1, 0, 0) \\ \therefore c_1 + i c_2 = 0 \quad \& \quad c_2 = 0 \Rightarrow c_1 = 0 \quad \therefore \underline{u}_1 \& \underline{u}_2 \text{ are Linearly Independent}$$

b) $\underline{u}_i$'s do not form an orthogonal basis [Inner Product $\neq 0$]

An orthogonal basis can always be obtained from a non-orthogonal basis.

c) Let $\underline{v}_i$'s form an Orthogonal basis. $\underline{u}_1$ is chosen as Reference, $\underline{u}_2$ needs to be re-oriented. $\underline{x}_i$'s would be corresponding orthonormal vectors.

$$\underline{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad \underline{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \& \quad \underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad \sqrt{\langle \underline{u}_1, \underline{u}_1 \rangle} = \|\underline{u}_1\| \text{ (Norm)} \\ \underline{u}_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}; \quad \text{Now } \underline{v}_2 = \underline{u}_2 - \alpha \underline{x}_1; \quad \langle \underline{v}_2, \underline{x}_1 \rangle = 0$$



$$\Rightarrow \left(\begin{bmatrix} i \\ 1 \end{bmatrix} - \alpha \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0 \Rightarrow i - \alpha = 0 \\ \therefore \alpha = i$$

$$\text{So, } \underline{v}_2 = \underline{u}_2 - i \underline{x}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix} - \begin{bmatrix} i \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (\text{Coincidentally Normalised too})$$

$$\text{Thus, } \underline{v}_1 = \underline{x}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \& \quad \underline{v}_2 = \underline{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

This Process is called Gram-Schmidt Orthogonalisation.

Qp. Determine Orthonormal Basis from $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ & $\begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}$

$$\underline{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \underline{v}_1; \quad \underline{x}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}; \quad \underline{u}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\text{Now, } \underline{v}_2 = \underline{u}_2 - \alpha \underline{x}_1 \quad \& \quad \langle \underline{v}_2, \underline{x}_1 \rangle = 0 \\ \left(\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} - \alpha \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \right) \cdot \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} = 0 \\ \frac{1}{\sqrt{2}} - \frac{\alpha}{2} - \frac{1}{\sqrt{2}} - \frac{\alpha}{2} = 0 \quad \therefore \alpha = 0 \Rightarrow \underline{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \& \quad \underline{x}_2 = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ -1/\sqrt{2} \end{bmatrix}$$

$$\underline{u}_3 = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix}; \quad \underline{v}_3 = \underline{u}_3 - \beta \underline{x}_2 - \gamma \underline{x}_1 \quad \& \quad \langle \underline{v}_3, \underline{x}_2 \rangle = 0, \quad \langle \underline{v}_3, \underline{x}_1 \rangle = 0$$

$$\text{Now, } \underline{v}_3 = \begin{bmatrix} -(\beta+\gamma)/\sqrt{2} \\ 3 \\ 4 + (\beta-\gamma)/\sqrt{2} \end{bmatrix}; \quad \text{Thus} \rightarrow \frac{-(\beta+\gamma)}{2} - \frac{2\sqrt{2}}{2} - \frac{(\beta-\gamma)}{2} = 0 \Rightarrow \beta = -2\sqrt{2} \\ \& \quad \frac{-(\beta+\gamma)}{2} + \frac{2\sqrt{2}}{2} + \frac{(\beta-\gamma)}{2} = 0 \Rightarrow \gamma = 2\sqrt{2}$$

$$\text{Thus, } \underline{v}_3 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} \quad \& \quad \underline{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Using Positive Definiteness of Inner Products :- $\langle \underline{u} + \alpha \underline{v}, \underline{u} + \alpha \underline{v} \rangle \geq 0$
 for every scalar $\alpha \in \mathbb{R}$
 $\Rightarrow \langle \underline{u}, \underline{u} \rangle + 2\alpha \langle \underline{u}, \underline{v} \rangle + \alpha^2 \langle \underline{v}, \underline{v} \rangle \geq 0$

$$\alpha_{\min} = \frac{-\langle \underline{u}, \underline{v} \rangle}{\langle \underline{v}, \underline{v} \rangle} \Rightarrow \langle \underline{u}, \underline{u} \rangle - \frac{\langle \underline{u}, \underline{v} \rangle^2}{\langle \underline{v}, \underline{v} \rangle} \geq 0$$

Thus, $\langle \underline{u}, \underline{u} \rangle \langle \underline{v}, \underline{v} \rangle \geq \langle \underline{u}, \underline{v} \rangle^2 \Rightarrow \|\underline{u}\| \|\underline{v}\| \geq |\langle \underline{u}, \underline{v} \rangle|$
 This is called the Cauchy - Buryakowski - Schwarz (CBS) inequality.

Now, consider $\rightarrow \|\underline{u} + \underline{v}\|^2 = \langle \underline{u} + \underline{v}, \underline{u} + \underline{v} \rangle = \|\underline{u}\|^2 + \langle \underline{u}, \underline{v} \rangle + \langle \underline{v}, \underline{u} \rangle + \|\underline{v}\|^2$
 $= \|\underline{u}\|^2 + \|\underline{v}\|^2 + 2 \operatorname{Re}(\langle \underline{u}, \underline{v} \rangle)$

But from CBS inequality :- $\|\underline{u}\| \|\underline{v}\| \geq \operatorname{Re}(\langle \underline{u}, \underline{v} \rangle)$

$$\therefore \|\underline{u}\|^2 + \|\underline{v}\|^2 + 2\|\underline{u}\| \|\underline{v}\| \geq \|\underline{u} + \underline{v}\|^2 \Rightarrow (\|\underline{u}\| + \|\underline{v}\|)^2 \geq \|\underline{u} + \underline{v}\|^2$$

$$\Rightarrow \|\underline{u}\| + \|\underline{v}\| \geq \|\underline{u} + \underline{v}\| \quad \text{This is the famous triangle inequality.}$$

Metric Space :-

A space is said to be a Metric Space if :-

- 1) Positivity - $d(\underline{u}, \underline{v}) \geq 0$
- 2) Symmetry - $d(\underline{u}, \underline{v}) = d(\underline{v}, \underline{u})$

- 3) Triangle inequality - $d(\underline{u}, \underline{v}) \leq d(\underline{u}, \underline{w}) + d(\underline{w}, \underline{v})$

A more general definition of p-Metric space $\rightarrow d_p(\underline{u}, \underline{v}) = \left[\sum_{i=1}^n |v_i - u_i|^p \right]^{\frac{1}{p}}$
 For Continuous functions $\rightarrow d_p(f, g) = \left[\int_a^b |f(x) - g(x)|^p dx \right]^{\frac{1}{p}}$

Q. A 3D space of polynomial functions of the form x^n & $n \in \mathbb{Z}^+$ is to be expanded using an orthonormal basis. Determine the basis.

$u_1(x) = 1$; $u_2(x) = x$; $u_3(x) = x^2$; Set $v_1(x) = 1$
 Now, $\|v_1(x)\| = \left(\int_{-1}^1 (v_1(x))^2 dx \right)^{\frac{1}{2}} = \left(x \Big|_{-1}^1 \right)^{\frac{1}{2}} = \sqrt{2}$

$\therefore w_1(x) = \frac{v_1(x)}{\|v_1(x)\|} = \frac{1}{\sqrt{2}} \rightarrow \text{Normalized}$; $v_2(x) = u_2(x) - \alpha v_1(x)$
 & $\langle v_1, v_1 \rangle = 0$

So, $v_2(x) = x - \alpha \Rightarrow \int_{-1}^1 (x - \alpha) \cdot 1 dx = 0$; $\alpha = 0$

$\therefore v_2(x) = u_2(x) = x$; $\|v_2(x)\| = \sqrt{\int_{-1}^1 x^2 dx} = \sqrt{\frac{2}{3}}$

So, $w_2(x) = \sqrt{\frac{3}{2}} x$; $v_3(x) = u_3(x) - \beta v_1(x) - \gamma v_2(x)$
 $= x^2 - \beta - \gamma x$

$$\text{Also, } \langle v_3, v_2 \rangle = 0 \quad \& \quad \langle v_3, v_1 \rangle = 0 \Rightarrow \int_{-1}^1 (x^2 - \gamma x - \beta) dx = 0$$

$$\& \quad \int_{-1}^1 (x^2 - \gamma x - \beta) x dx = 0$$

$$\Rightarrow \int_{-1}^1 \frac{x^3}{3} - \beta x - \gamma \frac{x^2}{2} dx = 0 \Rightarrow \frac{2}{3} - 2\beta = 0 \quad ; \quad \beta = \frac{1}{3}$$

$$\& \quad \int_{-1}^1 \frac{x^4}{4} - \beta x^2 - \gamma \frac{x^3}{3} dx = 0 \Rightarrow \gamma = 0 \quad ; \quad \therefore v_3(x) = x^2 - \frac{1}{3}$$

$$\|v_3\| = \sqrt{\int_{-1}^1 \left(x^2 - \frac{1}{3}\right)^2 dx} = \frac{2}{3} \sqrt{\frac{2}{5}} \quad ; \quad w_3(x) = \frac{3}{2} \sqrt{\frac{5}{2}} \left(x^2 - \frac{1}{3}\right)$$

L^2 space :-

Consider a space $X[a, b]$ of functions defined in the interval $[a, b]$ such that $f: [a, b] \rightarrow \mathbb{C}$ and $\rightarrow$

$\int_a^b |f(x)|^2 dx < \infty$; In such a case, the space is called L^2 space.

Qp. $L^2(0, a)$: a.) $f(x) = \sin(x)$ & $a = \pi$

$$\int_0^\pi |\sin x|^2 dx = \int_0^\pi \sin^2 x dx = \frac{1}{2} \int_0^\pi (1 - \cos 2x) dx = \frac{1}{2} x \Big|_0^\pi = \frac{\pi}{2}$$

b.) $f(x) = \cos(x)$, $a = \pi$; $\int_0^\pi |\cos x|^2 dx = \int_0^\pi \cos^2 x dx = \int_0^\pi 1 dx - \frac{\pi}{2} = \frac{\pi}{2}$

c.) $f(x) = \frac{1}{1+x}$, $a \rightarrow \infty$; $\int_0^\infty \frac{1}{(1+x)^2} dx = \left. -\frac{1}{(1+x)} \right|_0^\infty = 1$

Qp. 2.) $V = \{ e^{inx} \in \mathbb{R} \} ; L^2(-\pi, \pi)$

$$\int_{-\pi}^\pi (e^{inx})^2 dx = \int_{-\pi}^\pi e^{2inx} dx = \left. \frac{e^{2inx}}{2in} \right|_{-\pi}^\pi = \frac{e^{2ni\pi}}{2in} - \frac{e^{-2ni\pi}}{2in} = 0$$

[Non Normalisable]

To check for orthogonality : $\int_{-\pi}^\pi e^{inx} e^{imx} dx = \left. \frac{e^{i(m+n)x}}{i(m+n)} \right|_{-\pi}^\pi$

$$= \frac{1}{i(m+n)} \times (e^{i\pi(m+n)} - e^{-i\pi(m+n)}) = \frac{\sin((m+n)\pi)}{m+n}$$

If it is 0 for $m+n \neq 0$, then orthogonal

Adjoint of an Operator

$$\text{If } L = \frac{d^2}{dx^2} + x \frac{d}{dx}$$

$$\begin{aligned} & \text{; } x=0, \quad du/dx = 0 \\ & \quad x=1, \quad du/dx + u = 0 \end{aligned}$$

Find L^* .

$$\underline{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \underline{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \Rightarrow \underline{A} \underline{u} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} a_{11}u_1 + a_{12}u_2 \\ a_{21}u_1 + a_{22}u_2 \end{bmatrix}$$

$$\text{Let } \underline{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}; \quad \text{Now } \langle \underline{A} \underline{u}, \underline{v} \rangle = a_{11}u_1v_1 + a_{12}u_2v_1 + a_{21}u_1v_2 + a_{22}u_2v_2 \\ = u_1(a_{11}v_1 + a_{21}v_2) + u_2(a_{12}v_1 + a_{22}v_2)$$

$$\therefore \langle \underline{A} \underline{u}, \underline{v} \rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11}v_1 + a_{21}v_2 \\ a_{12}v_1 + a_{22}v_2 \end{bmatrix} \right\rangle = \left\langle \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \right\rangle$$

Thus, $\langle \underline{A} \underline{u}, \underline{v} \rangle = \langle \underline{u}, \underline{B} \underline{v} \rangle$; B is Adjoint for Operator A .

Q. Find Adjoint operator for $\underline{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$

$$\text{Let } \underline{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \text{ \& } \underline{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}; \quad \underline{A} \underline{u} = \begin{bmatrix} a_{11}u_1 + a_{12}u_2 + a_{13}u_3 \\ a_{21}u_1 + a_{22}u_2 + a_{23}u_3 \end{bmatrix}$$

$$\begin{aligned} \langle \underline{A} \underline{u}, \underline{v} \rangle &= a_{11}u_1v_1 + a_{12}u_2v_1 + a_{13}u_3v_1 + a_{21}u_1v_2 + a_{22}u_2v_2 + a_{23}u_3v_2 \\ &= u_1[a_{11}v_1 + a_{21}v_2] + u_2[a_{12}v_1 + a_{22}v_2] + u_3[a_{13}v_1 + a_{23}v_2] \\ &= \left\langle \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \underline{B} \underline{v} \right\rangle, \text{ where } \underline{B} = \begin{bmatrix} a_{11} & a_{21} \\ a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix} \end{aligned}$$

B would be Adjoint Transform, Not Operator

Note :- 1.) The Eigenvalues of self Adjoint operator are always real.

$$\langle \underline{A} \underline{u}, \underline{u} \rangle = \langle \lambda \underline{u}, \underline{u} \rangle = \lambda \langle \underline{u}, \underline{u} \rangle; \quad \langle \underline{u}, \underline{A} \underline{u} \rangle = \langle \underline{u}, \lambda \underline{u} \rangle = \bar{\lambda} \langle \underline{u}, \underline{u} \rangle$$

$$\therefore \lambda = \bar{\lambda} \Rightarrow \lambda \text{ is always Real.}$$

2.) The Eigenvectors of a self Adjoint operator corresponding to distinct Eigenvalues are always Orthogonal.

$$\begin{aligned} \lambda_1 &\rightarrow v_1; \quad \lambda_2 \rightarrow v_2 \\ \langle \underline{A} v_1, v_2 \rangle &= \langle \lambda_1 v_1, v_2 \rangle = \lambda_1 \langle v_1, v_2 \rangle \\ \langle v_1, \underline{A} v_2 \rangle &= \langle v_1, \lambda_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \end{aligned}$$

Both equal $\Rightarrow (\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle = 0$

Fredholm's Alternative Theorem

Q. Determine the Range Space of the following system of equations $\rightarrow$
 $x_1 + x_2 + x_3 = 1$, $2x_1 + 3x_2 + x_3 = 2$, $5x_1 + 6x_2 + 4x_3 = 1$

$$\begin{bmatrix} 1 & 1 & 1 & | & b_1 \\ 2 & 3 & 1 & | & b_2 \\ 5 & 6 & 4 & | & b_3 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 1 & 1 & | & b_1 \\ 0 & 1 & -1 & | & b_2 - 2b_1 \\ 0 & 1 & -1 & | & b_3 - 5b_1 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 1 & 1 & | & b_1 \\ 0 & 1 & -1 & | & b_2 - b_1 \\ 0 & 0 & 0 & | & b_3 - b_2 - 3b_1 \end{bmatrix}$$

$$\therefore b_3 - b_2 - 3b_1 = 0 \Rightarrow b_1 = \alpha \quad \& \quad b_2 = \beta$$

$$\therefore b_3 = 3\alpha + \beta$$

$$\text{So, } \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$\underline{A} \underline{x} = \underline{b}$ is solvable if $\rightarrow \langle \underline{b}, \underline{y} \rangle = 0 : \underline{A}^* \underline{y} = 0$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 5 & 6 & 4 \end{bmatrix} \Rightarrow A^* = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 3 & 6 \\ 1 & 1 & 4 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 5 & | & 0 \\ 1 & 3 & 6 & | & 0 \\ 1 & 1 & 4 & | & 0 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 1 & 2 & 5 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & -1 & -1 & | & 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1 & 2 & 5 & | & 0 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\text{Set } y_2 = \alpha \Rightarrow y_3 = -\alpha \quad \& \quad y_1 = 3\alpha$$

$$\therefore \underline{y} = \alpha \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}; \quad \langle \underline{b}, \underline{y} \rangle = \alpha (3b_1 + b_2 - b_3) = 0$$

$$\therefore 3b_1 + b_2 - b_3 = 0 \Rightarrow b_1 = \alpha; \quad b_2 = \beta \quad \& \quad b_3 = 3\alpha + \beta$$

$$\text{So, } \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} + \beta \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \rightarrow \text{Same Answer}$$

Eigenvalue Problems

$$a) \frac{dx}{dt} = 2x$$

$$b) \frac{dx}{dt} = x + 2y$$

$$\frac{dy}{dt} = -3y$$

$$\frac{dy}{dt} = y$$

$$a) \frac{dx}{dt} = 2x \Rightarrow \int \frac{dx}{x} = 2 \int dt \Rightarrow x = C_1 e^{2t} \quad \& \quad \frac{dy}{dt} = -3y \Rightarrow \int \frac{dy}{y} = -3 \int dt \Rightarrow y = C_2 e^{-3t}$$

Independent / Coupled Equations

$$b) \frac{dy}{dt} = y \Rightarrow y = C_1 e^t; \therefore \frac{dx}{dt} - x = 2C_1 e^t \quad (I.F. = e^{-t} = e^{-\int 1 dt})$$

$$\Rightarrow e^{-t} \frac{dx}{dt} - x e^{-t} = 2C_1 \Rightarrow \int d(x e^{-t}) = 2C_1 \int dt \Rightarrow x = (2C_1 t + C_2) e^t$$

So, $\left. \begin{array}{l} \frac{dx}{dt} = x + 2y \\ \frac{dy}{dt} = x \end{array} \right\} \Rightarrow \frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow \frac{d\underline{X}}{dt} = \underline{A} \underline{X}$ Now an Eigenvalue Problem

$$\begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 1 & -\lambda \end{bmatrix} \Rightarrow -\lambda + \lambda^2 - 2 = 0 \quad \text{So, } \lambda_1 = -1 \Rightarrow v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore \lambda = -1, 2 \quad \lambda_2 = 2 \Rightarrow v_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = - \begin{bmatrix} x \\ y \end{bmatrix} \Rightarrow x + 2y = -x \text{ \& } -x = y \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = k_1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\& \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 2y \end{bmatrix} \Rightarrow x + 2y = 2x \text{ \& } x = 2y \Rightarrow \begin{bmatrix} x \\ y \end{bmatrix} = k_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\underline{X} = C_1 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + C_2 e^{2t} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{Thus, } \begin{array}{l} x(t) = C_1 e^{-t} + 2C_2 e^{2t} \\ y(t) = -C_1 e^{-t} + C_2 e^{2t} \end{array}$$

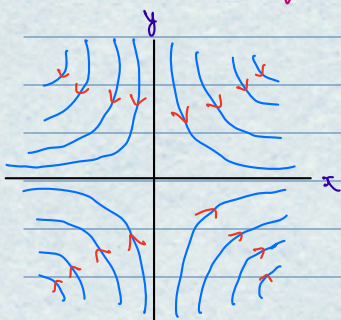
In general, $\frac{d\underline{X}}{dt} = \underline{A} \underline{X} \Rightarrow \underline{X}(t) = \sum_{i=1}^n C_i e^{\lambda_i t} \underline{v}_i$

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}; \text{ Eigenvalues } \rightarrow \lambda_1 = 2 \text{ \& } \lambda_2 = -3$$

$$\text{Eigenvectors } \rightarrow \underline{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ \& } \underline{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x \\ y \end{bmatrix} = C_1 e^{2t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + C_2 e^{-3t} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{array}{l} x(t) = C_1 e^{2t} \\ y(t) = C_2 e^{-3t} \end{array} \left. \vphantom{\begin{array}{l} x(t) \\ y(t) \end{array}} \right\} \text{ same solution as before}$$

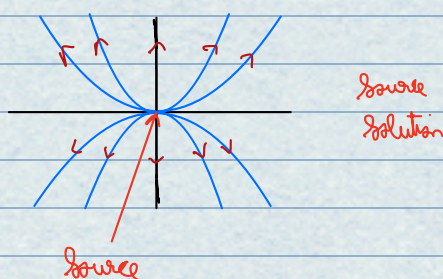
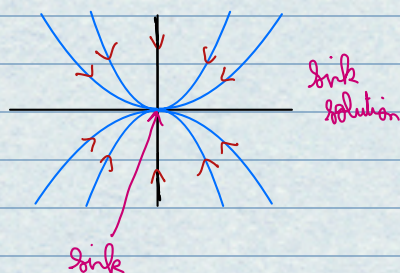
Simultaneous Equations means their solution must be true together.



For this problem, the coordinate axes are also eigenvectors (Arrows indicate time evolution)

Saddle solution: However, if both Eigenvalues are +ve, both exponentials blow up. If both are -ve, exponentials die down.

Linear stability Analysis



a) $\frac{d^2 y}{dx^2} - y = 0$

b) $\frac{d^2 y}{dx^2} + y = 0$

$$\text{Let } \frac{dy}{dx} = z \Rightarrow \frac{d}{dx} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\lambda_1 = 1 \quad \underline{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \& \quad \underline{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} y \\ z \end{bmatrix} = C_1 e^x \begin{bmatrix} 1 \\ 1 \end{bmatrix} + C_2 e^{-x} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow y = C_1 e^x + C_2 e^{-x} \quad \& \quad \frac{dy}{dx} = C_1 e^x - C_2 e^{-x}$$

Saddle Solution

$$\frac{d}{dx} \begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\lambda_1 = i \quad \& \quad \lambda_2 = -i \quad ; \quad \underline{v}_1 = \begin{bmatrix} -i \\ 1 \end{bmatrix} \quad \& \quad \underline{v}_2 = \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} y \\ z \end{bmatrix} = C_1 e^{ix} \begin{bmatrix} -i \\ 1 \end{bmatrix} + C_2 e^{-ix} \begin{bmatrix} i \\ 1 \end{bmatrix}$$

$$\Rightarrow y = -C_1 i e^{ix} + C_2 i e^{-ix}$$

$$\Rightarrow y = -C_1 i (\cos x + i \sin x) + C_2 i (\cos x - i \sin x)$$

$$\therefore y = (C_1 + C_2) \sin x + (C_2 - C_1) i \cos x$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} (C_1 + C_2) \sin x \\ (C_1 + C_2) \cos x \end{bmatrix} + i \begin{bmatrix} (C_2 - C_1) \cos x \\ (C_1 - C_2) \sin x \end{bmatrix}$$

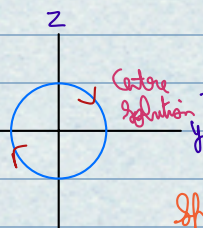
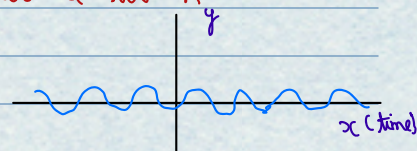
$$= z_1 + i z_2 \quad (\text{Let})$$

$$\therefore \frac{d^2}{dx^2} (z_1 + i z_2) + (z_1 + i z_2) = 0 \Rightarrow \left(\frac{d^2}{dx^2} z_1 + z_1 \right) + i \left(\frac{d^2}{dx^2} z_2 + z_2 \right) = 0$$

Real part is a solution, Imaginary Part (without i) is also a solution.

$$\therefore y = (C_1 + C_2) \sin x + i(C_2 - C_1) \cos x \Rightarrow y = A \sin(x) + B \cos x$$

(solution)

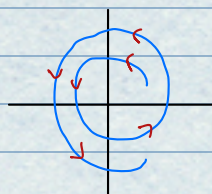


Periodic / Oscillatory Behaviour (For imaginary Eigenvalues) [Non-Monotonic Behaviour]

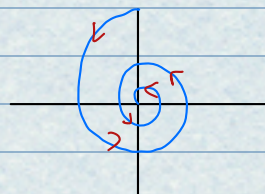
★ If Eigenvalues are Complex [Non Zero Real & Imaginary Parts], then spiral Plots are observed.

Eg:- ① λ_s are $(2+i)$ & $(2-i)$, then spiral source solution [+ve real part causes divergence.]

② λ_s are $(-2+i)$ & $(-2-i)$, then spiral sink solution [-ve real part causes convergence.]



spiral source solution



spiral sink solution

Similarity Transformation

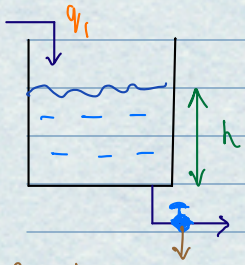
$$2) \frac{dx}{dt} = x + y - t - 1$$

$$\frac{dy}{dt} = 4x + y - t - 2$$

Solution: $\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y-t-1 \\ 4x+y-4t+2 \end{bmatrix}$ Clearly not of the form $\frac{d\underline{x}}{dt} = \underline{A}\underline{x}$

$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}}_{\text{Autonomous part}} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} -t-1 \\ -4t+2 \end{bmatrix}$ contains independent variable t $\Rightarrow \frac{d\underline{x}}{dt} = \underline{A}\underline{x} + b(t)$

Note: - For $\frac{d\underline{x}}{dt} = \underline{A}\underline{x}$, $\underline{x} = \sum_{i=1}^n c_i e^{\lambda_i t} \underline{v}_i$ Autonomous systems



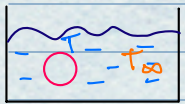
Only drainage $\Rightarrow q_1 = 0$; $\frac{dh}{dt} = \frac{1}{A} (q_1 - q_2) = -\frac{q_2}{A}$; $\therefore \frac{dh}{dt} = -\frac{b(h)}{A}$

Autonomous systems govern themselves

If now a pump is added at the outlet, $\frac{dh}{dt} = -\frac{q_2}{A} - b_2(t)$

gravity drainage: $\frac{dh}{dt} = \frac{1}{A} C_d \sqrt{2gh}$

Can be governed by external parties at will. (Non Autonomous)



$\frac{dT}{dt} = \alpha (T - T_\infty)$ Autonomous system

$\hookrightarrow$ if $T_\infty = T_\infty(t)$ by nozzles, etc. then Non Autonomous

$\underline{A} = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$; $\lambda_1 + \lambda_2 = 2$ & $\lambda_1 \lambda_2 = 1 - 4 = -3 \Rightarrow \lambda_1 = -1$ & $\lambda_2 = 3$
 $\therefore \underline{v}_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$ & $\underline{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\underline{P}$ = Augmented Matrix of Eigenvectors $= \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix}$; $\underline{P}^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix}$

$\Rightarrow \underline{P}^{-1} \underline{A} = \frac{1}{4} \begin{bmatrix} 2 & -1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -2 & 1 \\ 6 & 3 \end{bmatrix}$

$\Rightarrow \underline{P}^{-1} \underline{A} \underline{P} = \frac{1}{4} \begin{bmatrix} -2 & 1 \\ 6 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -2 & 2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -4 & 0 \\ 0 & 12 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$

This is a Diagonal Matrix, with Eigenvalues in the Diagonal

$\underline{A}$ & $\underline{P}^{-1} \underline{A} \underline{P}$ are similar Matrices (same Eigenvalues); But different Eigenvectors

* Scalar Analysis of the Problem: $\frac{dx}{dt} = ax + b(t) \Rightarrow \frac{dx}{dt} - ax = b(t)$

This is a Linear Differential Equation; Integrating Factor $\rightarrow e^{\int -a dt} = e^{-at}$

$\therefore \frac{d}{dt} [e^{-at} x] = e^{-at} b(t) \Rightarrow e^{-at} x = \int e^{-at} b(t) dt + C$

$\Rightarrow x(t) = e^{at} + e^{at} \int e^{-at} b(t) dt$

Spring to the Matrix Equation:-

$$\frac{d}{dt} \underline{x} - \underline{A} \underline{x} = \underline{b}(t) \Rightarrow \frac{d}{dt} (\underline{P}^{-1} \underline{x}) - \underline{P}^{-1} \underline{A} \underline{x} = \underline{P}^{-1} \underline{b}(t)$$

$$\Rightarrow \frac{d}{dt} (\underline{P}^{-1} \underline{x}) - (\underline{P}^{-1} \underline{A} \underline{P}) (\underline{P}^{-1} \underline{x}) = \underline{P}^{-1} \underline{b}(t) ; \text{ let } \underline{P}^{-1} \underline{x} = \underline{y}$$

$$\& \underline{P}^{-1} \underline{A} \underline{P} = \underline{\Delta} \text{ (Diagonal Matrix)} \cdot \text{let } \underline{f}(t) = \underline{P}^{-1} \underline{b}(t)$$

$$\text{Then, } \frac{d}{dt} \underline{y} - \underline{\Delta} \underline{y} = \underline{f}(t) ; \text{ This has now become a Linear O.E.}$$

$$\text{Integrating Factor: } e^{-\int \underline{\Delta} dt} = e^{-\underline{\Delta} t} = e^{-t \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}} \quad \left[\begin{matrix} \text{Matrix} \\ e \end{matrix} \right]$$

$$\text{Now, } e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad \text{(Taylor Expansion)} \quad -\underline{\Delta} t = \begin{bmatrix} t & 0 \\ 0 & -3t \end{bmatrix}$$

$$\Rightarrow e^{-\underline{\Delta} t} = I - \underline{\Delta} t + \frac{1}{2!} (-\underline{\Delta} t)^2 + \dots$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} t & 0 \\ 0 & -3t \end{bmatrix} + \frac{1}{2!} \begin{bmatrix} t^2 & 0 \\ 0 & 9t^2 \end{bmatrix} + \dots$$

$$= \begin{bmatrix} e^t & 0 \\ 0 & e^{-3t} \end{bmatrix}$$

$$\text{Thus, } \frac{d}{dt} (e^{-\underline{\Delta} t} \underline{y}) = \underline{f}(t) e^{-\underline{\Delta} t} \Rightarrow \underline{y}(t) = e^{\underline{\Delta} t} \underline{C} + e^{\underline{\Delta} t} \int e^{-\underline{\Delta} t} \underline{f}(t) dt$$

$$\text{So, } \underline{x}(t) = \underline{P} \left[e^{\underline{\Delta} t} \underline{C} + e^{\underline{\Delta} t} \int e^{-\underline{\Delta} t} \underline{f}(t) dt \right] \quad \text{Where } \underline{\Delta} = \underline{P}^{-1} \underline{A} \underline{P}$$

Sturm-Liouville Theory: Bessel Equation - Frobenius Method

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2) y = 0 ; n \geq 0 ; \text{ let } y = \sum_{i=0}^{\infty} a_i x^{i+k}, \frac{dy}{dx} = \sum_{i=0}^{\infty} a_i (i+k) x^{i+k-1}$$

$$\& \frac{d^2 y}{dx^2} = \sum_{i=0}^{\infty} a_i (i+k)(i+k-1) x^{i+k-2} ; \text{ so } \rightarrow \sum_{i=0}^{\infty} a_i (i+k)(i+k-1) x^{i+k} + \sum_{i=0}^{\infty} a_i (i+k) x^{i+k}$$

$$- n^2 \sum_{i=0}^{\infty} a_i x^{i+k} + \sum_{i=0}^{\infty} a_i x^{i+k+2} = 0$$

$$\therefore \sum_{i=0}^{\infty} a_i [(i+k)^2 - n^2] x^{i+k} + \sum_{i=0}^{\infty} a_i x^{i+k+2} = 0$$

Polynomial = 0
 $\Rightarrow$ Every Coefficient = 0

This series lagging by 2 powers $\xrightarrow{\hspace{1cm}}$ This series ahead by 2 powers

$$\begin{aligned} \text{Coefficient of } x^k: & a_0 (k^2 - n^2) x^k = 0 \Rightarrow k = \pm n ; a_0 \neq 0 \\ \text{Coefficient of } x^{k+1}: & a_1 ((k+1)^2 - n^2) = 0 \Rightarrow k = -1 \pm n \text{ or } (a_1 = 0?) \end{aligned}$$

$$\text{Solving for } k = \pm n \& a_1 = 0 \rightarrow \sum_{i=0}^{\infty} a_i [(i+n)^2 - n^2] x^{i+n} + \sum_{i=0}^{\infty} a_i x^{i+n+2} = 0$$

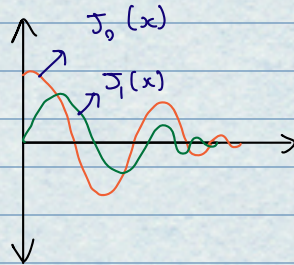
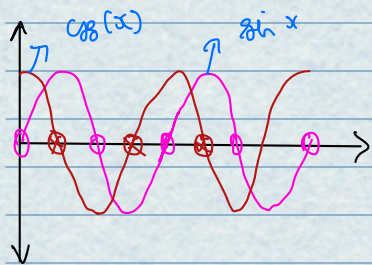
$$\text{Coefficient of } x^{n+2}: a_2 [(n+2)^2 - n^2] + a_0 = 0 \Rightarrow a_2 = \frac{-a_0}{(4n+4)}$$

$$a_3 = \frac{-a_1}{(n+3)^2 - n^2} = 0 \quad [As \ a_1 = 0] ; \quad a_4 = \frac{-a_2}{(n+4)^2 - n^2} = \frac{a_0}{32(n+1)(n+2)}$$

$$a_5 = a_7 = a_9 \dots = 0 \quad \text{Half of the series killed}$$

$$a_6 = \frac{-a_4}{(n+6)^2 - n^2} = \frac{-a_0}{32 \times 12 (n+1)(n+2)(n+3)} ; \text{ We have effectively only 1 unknown, } a_0$$

A specific value of a_0 gives a clean solution : $a_n = \frac{1}{2^n \times n!}$



$J_1, J_2, \dots, J_n \rightarrow$ Linearly Independent & Orthogonal

Bessel Function of Second Kind $\rightarrow$ Obtained by transformation on Bessel Function of 1st order

Note :- Sturm-Liouville Operator, $\hat{L} = \frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x)$; This operator constitutes

Self Adjoint Problem, with Real Eigen values